Udayan Atreya

San Jose, CA | 669-340-6033 | udayan.atreya@gmail.com | LinkedIn - /uatreya | Github | Portfolio

EDUCATION

Master of Science, Computer Science May 2026
San Jose State University, San Jose, US
Coursework: Multimodal ML, Retrieval Augmented Generation (RAG), Applied Deep Learning, Biometric Security with AI.

Bachelor of Technology, Computer Science & Engineering May 2020
Manipal University, Jaipur, India
Coursework: Data Structures & Algorithms, Relational Database Management Systems, Object-Oriented Programming.

TECHNICAL SKILLS

AI/ML Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, RAGAS, Whisper (ASR)
LLMs, Multimodal & Generative AI: GPT-4o, Gemini, Llama-3, Claude, DeepSeek, Groq, Ollama, Vapi, Eleven Labs
AI Agent Systems: Agentic RAG, reasoning & planning pipelines, tool use, multi-agent collaboration, semantic routing
MLOps & Cloud: AWS (Lambda, S3, API Gateway, DynamoDB), CI/CD, Prometheus, Grafana, FastAPI, MongoDB, Pinecone, ChromaDB
Languages: Python, Java, C#, SQL, JavaScript

PROJECT EXPERIENCE

Evaluator-Optimizer Agentic RAG *(LangChain, GPT-4o, Gemini, Llama-3, ChromaDB, Whisper, PyTorch)* [GitHub]
Aug 2025 – Mar 2026

- Architected a multimodal AI agent system with Evaluator-Optimizer feedback loop for Language-Based Video Reasoning (LBVR), processing audio (Whisper ASR), text, and video modalities; **reduced hallucination by 25%** and **improved answer accuracy by 18%** over baselines.
- Implemented reasoning and planning pipelines with tool-use capabilities and multi-agent collaboration enabling iterative self-refinement across heterogeneous knowledge sources.
- Engineered production-grade RAG evaluation suite using RAGAS and Llama-3-70b (LLM-as-judge) over 500+ STEM QA pairs, quantifying Faithfulness, Answer Relevancy, Context Precision, Noise Sensitivity, and Semantic Similarity for end-to-end ML monitoring.
- Optimized latency-accuracy trade-off via Dynamic Chunking and MMR search, **reducing context redundancy by 40%** while maintaining **90%+ semantic coverage**; ablation study showed lightweight MiniLM at **98% parity** with SOTA BGE-base, **cutting storage costs by 50%**.

Agentic BMC Observability Platform *(LangChain, Gemini 1.5, FastAPI, MongoDB, AWS S3)* [GitHub]
Jun 2025 – Jul 2025

- Built a production-deployed AI agent system with semantic router enabling autonomous reasoning, planning, and tool use for infrastructure observability; achieved **95% query classification accuracy** and **30% reduction in manual triage time**.
- Designed multi-agent collaboration layer integrating Prometheus/Grafana telemetry with a hybrid retrieval engine (MongoDB + S3), **reducing root-cause-analysis retrieval time by 75%** (2m to 30s) — demonstrating end-to-end ML pipeline deployment and monitoring.
- Enforced human-in-the-loop validation and role-based policy checks for agentic hardware interventions on Baseboard Management Controllers, **preventing 100% of unauthorized state changes**.

WORK EXPERIENCE

Software Engineer, Accenture, Bengaluru, India Dec 2021 – Jul 2024
Associate Software Engineer, Accenture, Bengaluru, India Nov 2020 – Nov 2021

- Developed an analytics-driven recommendation engine applying weighted scoring and NLP-based intent routing to personalize automation tool suggestions for users, **reducing discovery time by 30%** — demonstrating AI system integration into user-facing products.
- Partnered cross-functionally to deliver a **ChatOps platform** directing natural language queries to automation workflows via serverless microservices, **reducing overhead by 40%** (~5m to ~1m); integrated scalable API gateway for production traffic.
- Architected a user notification microservices system on **AWS (API Gateway, Lambda, DynamoDB)**, achieving a **20% increase in DAU**; optimized DynamoDB access patterns **reducing retrieval latency by 37%**.

- Delivered a serverless code vulnerability remediation pipeline (AWS Lambda, S3, EventBridge, Azure DevOps) safeguarding **400+ applications** with automated end-to-end deployment and monitoring, **cutting MTTR by 80%.**